

House Price Prediction – Project Summary

Project Overview

This project focused on building a machine learning model to predict house prices using different property features. The main objective was to understand how factors such as area, number of bathrooms, air conditioning, and other amenities affect the price of a house. Using Python and various data analysis libraries, I developed a model that could estimate house prices based on historical data..

Methodology

The workflow began with thorough data preparation: cleaning duplicate records, handling missing values, and encoding categorical data into numerical formats. Using an 80:20 train-test split, I compared two regression algorithms: Linear Regression and Random Forest Regressor. Performance was evaluated using MAE, RMSE, and R^2 scores.

Key Insights

Primary Drivers: Total area and the number of bathrooms are the most significant predictors of value.

Secondary Factors: Air conditioning and the number of stories notably increase property desirability and price.

Model Behavior: While effective for standard properties, the model tends to slightly underestimate luxury home values.

Performance

Among the two models, Linear Regression performed better and achieved an R^2 score of approximately 65%. The analysis showed that the area of the house had the strongest influence on price. Other important factors included the number of bathrooms, availability of air conditioning, and the number of stories. In general, larger houses with better amenities tended to have higher prices.

Conclusion

This project demonstrated how machine learning can be used to predict house prices and identify the factors that influence property value. The model provided reasonably accurate predictions and can serve as a useful tool for basic real estate price estimation. Through this project, I gained practical experience in data cleaning, data visualization, model building, and performance evaluation using Python.